TECHNICAL DESIGN DOCUMENT

Project Title: Integrated Solar Utility System for Sustainable Rural Communities (ISUS)

1. Introduction

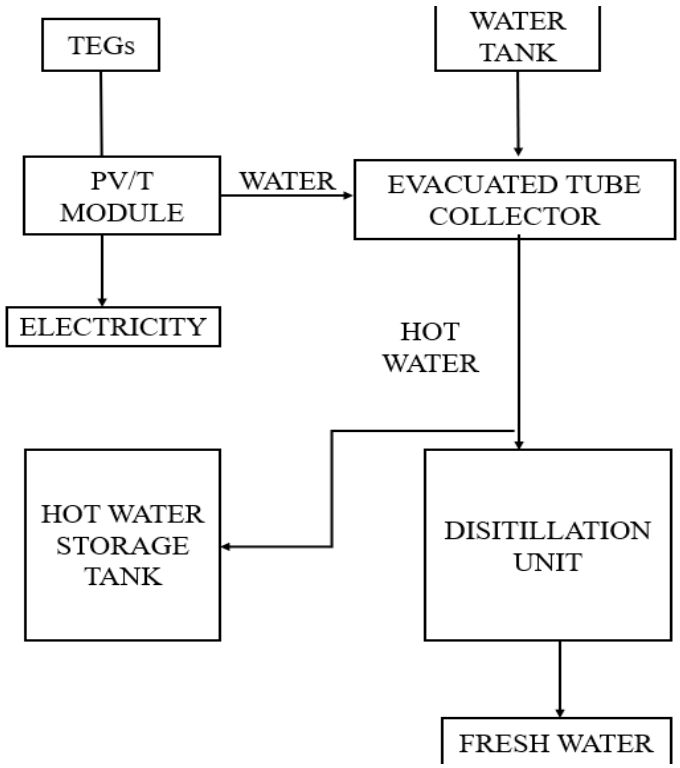
The Integrated Solar Utility System for Sustainable Rural Communities is designed to combine solar electricity generation, thermal energy collection, and water purification. The system integrates PV/T modules, TEGs, ETCs, and a solar distillation unit with PCM to support essential utility requirements in rural communities.

2. System Architecture

The system consists of the following major subsystems:

- Solar photovoltaic and thermal energy collection
- Thermoelectric power generation
- Electrical power conversion using buck and boost converters
- Solar thermal water heating using evacuated tube collectors

3. Architectural Diagram



3. Hardware Components and Specifications

- Solar panel: 20 W, 12 V
- Structure frame: Iron
- Metal sheets: 2 units
- PV/T module: Photovoltaic and thermal energy collection
- TEG modules: Convert thermal energy into electrical energy
- ETCs: Collect solar thermal energy for water heating
- Distillation unit: Produces distillate through evaporation and condensation
- PCM: Stores thermal energy beneath the distillation basin
- Buck converter: Reduces DC voltage
- Boost converter: Increases DC voltage

4. Working Principle

- The PV/T module collects solar energy and produces electrical and thermal energy. The TEG configuration converts a temperature difference into electrical energy through the Seebeck effect. The electrical output can be supplied to buck or boost converters depending on the required voltage.
- Water from the tank is routed through the thermal collection arrangement. The ETCs heat the water, which can be supplied for domestic hot-water applications and to the distillation unit. In the distillation unit, solar heat evaporates the water, and the vapour condenses to form distillate. PCM beneath the basin stores thermal energy and releases it as the temperature decreases.

5. TEG Simulation Parameters and Outputs

The reported simulation configuration uses four TEG modules with a Seebeck coefficient of 0.025 V/K, a hot-side temperature of 692 K, a cold-side temperature of 238 K, and a resistive load of 10 Ω.

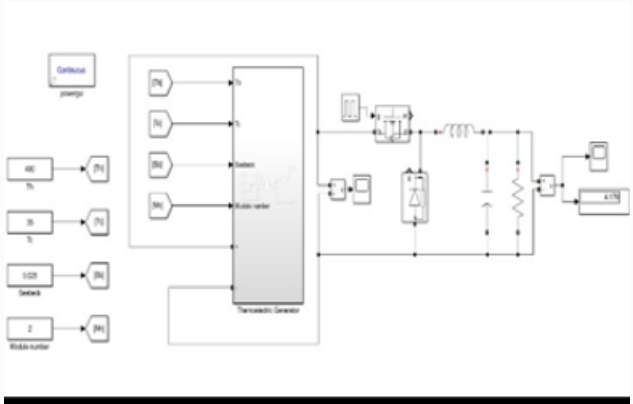
The reported TEG output voltage is 30 V.

For the boost converter, the input voltage is 30 V, the inductance is 1 mH, the capacitance is 470 µF, and the load resistance is 280 Ω. The reported output voltage is 37 V.

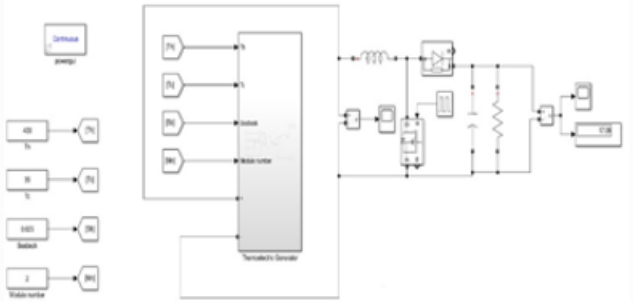
For the buck converter, the input voltage is 30 V, the inductance is 1 mH, the capacitance is 470 µF, and the load resistance is 32 Ω. The reported output voltage is 6 V.

These values represent the supplied simulation results and must be distinguished from experimental measurements.

6. Electrical Circuit Diagram



The thermoelectric generator is connected to the Buck converter



The thermoelectric generator is connected to the Boost

7. Mechanical Design



The proposed structure uses an iron frame and two metal sheets to support the components. The frame should provide sufficient mechanical stability for the solar panel, collectors, and distillation unit.

8. Safety Considerations

Electrical insulation, suitable fuses, secure mounting, safe water circulation, and protection against hot surfaces must be considered. The collected distillate must undergo appropriate water-quality testing before it is declared potable.

9. Performance Evaluation

System performance will be evaluated using solar-panel voltage and current, TEG output, converter output voltage, water inlet and outlet temperatures, PCM temperature, hot-water production, and daily distillate volume.

10. Safety Considerations

The system requires suitable electrical insulation, overcurrent protection, secure mechanical mounting, and safe handling of hot water and hot surfaces. The water circulation arrangement must be designed for safe operation. Collected distillate must undergo appropriate water-quality testing before it is considered suitable for drinking.